

# CHAPTER 1

## INTRODUCTION

### 1. THE FUNDAMENTAL QUESTIONS OF CONTROL

Imagine you are a child faced with the daunting task of acquiring the grammar of control in your language. You toddle around buoyantly (you should be past 3 by now), occasionally bumping into acoustic signals that miraculously map to “linguistic input”. Some of them sound like this:

- (1) a. Robin, do you want \_\_\_\_ to play with Kittie together?
- b. Come on, let me show you how \_\_\_\_ to feed her.
- c. No Robin, Kittie doesn’t like \_\_\_\_ to be smacked.
- d. Robin, look what you’ve done! Bad boy! Time \_\_\_\_ to go to bed.

From your shelter under the kitchen table, you may draw the following conclusions:

i) Mommy is very mad now; ii) Kittens make rotten toys; iii) My name must be Robin.

Apart from the lesson in parental control, you also ought to learn something about grammatical control. In each of the sentences above, an element is missing (from the underlined position) that is nonetheless “filled-in” by your target grammar. This is what linguists term the “understood subject” of the infinitive. In order to be able to understand such sentences and produce similar ones yourself, you have to figure out the reference of the understood subject in every case. Thus, unless you are after some big trouble with Mommy, you had better conclude that the understood subject is Robin and Mommy in (1a), Robin in (1b), Kittie in (1c) and everyone (especially Robin!) in (1d).

Those referential inferences are neither arbitrary nor fully deterministic; by and large, they are determined by your grammar, which also specifies where and how much room for indeterminacy exists. *Control* is the relation between some antecedent and the understood subject, or, to use current terminology, between the controller and PRO. Mastering the properties of this relation is the task of acquiring the grammar of control.

What does the child need to know about the control relation? Minimally, it seems that the following questions must be answered:

- (2) a. What elements/positions can control?  
 b. What elements/positions can be controlled?  
 c. What is the typology of control? (how many different types are there?)  
 d. Can the typology be deduced from principles of UG?  
 e. How is the controlled position (PRO) interpreted?  
 f. How is a controller picked up in a given structure?

All these issues must be resolved in order for the child to acquire full competence in the grammar of control. Quite literally, these are also the fundamental challenges the linguist faces in the study of control. Let us consider how this book addresses those challenges.

Question (2a) leads to an investigation of the syntactic underpinnings of control. In a substantial class of cases, the controller must stand in a certain structural relation to PRO (or the containing infinitive). Chapter 3 of this book analyses this relation and shows how it differentiates among various types of control. The issue of implicit arguments qua controllers also falls under this question; it is discussed in chapter 5.

Question (2b) concerns the distribution of PRO, and the answer to it is familiar - most probably, PRO occurs only in the subject position of non-finite clauses. This study has virtually nothing to add to this observation. The history of this problem, from the PRO-theorem (Chomsky 1981) until the null case analysis (Chomsky & Lasnik 1993), has never succeeded in deriving this fact without recourse to special stipulations; perhaps the distribution of PRO is an irreducible fact of UG.

Question (2c) is a major research question and has been at the center of lively debates within various grammatical frameworks. It also forms the methodological backbone of this book. One of my central claims is that the traditional typologies of control must be rethought and revised. Very broadly, the proposal is twofold: i) The category of Obligatory Control (OC) is not homogenous but rather consists of two subtypes - Exhaustive and Partial Control (EC and PC); ii) The line between OC and Non-Obligatory Control (NOC) is not located where previous accounts hold, and the key to the right line is the proper analysis of the Super-Equi construction. Chapters 2 and 3 of this book explore the typology of control in detail, and attempt to deduce it from general properties of syntactic relations (based on *Agree*), the projection of Tense and phi-features, the structure of VP-shells and interface conditions, approaching the goal set in (2d).

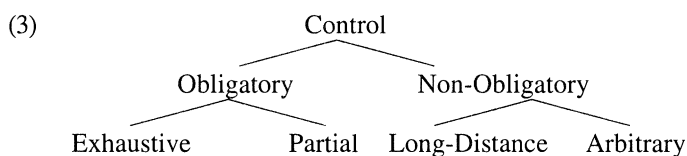
This study provides a partial answer to (2e), in various domains. As discussed in chapter 2, PRO in OC may be interpreted as either identical to or inclusive of the controller, an option related to the interaction of semantic number with control. In chapter 5 we argue that in both cases, OC is *not* interpreted via a predication mechanism, so that PRO does not function as a lambda-variable; and chapter 3 shows that PRO in NOC is interpreted as a logophor.

Finally, controller choice, the subject matter of question (2f), lies at the crossroads of syntax, semantics and pragmatics. As regards OC, syntax delineates the search domain but does not force a particular choice within that domain. Chapter 5 discusses the various semantic and pragmatic considerations affecting the choice (and/or shifting) of the controller, a complex topic with as yet many unsolved problems.

Let us turn now to brief description of the major proposals of this study.

## 2. A TYPOLOGY OF CONTROL

The general typology I will advance is the following:



The nodes in this diagram are labeled according to standard practice, whereby the referential properties of PRO define the relevant type of control. The terminology itself does not carry any theoretical commitment; however, the partition it represents incorporates a substantial empirical hypothesis. That is, we demonstrate that the empirical data cluster neatly under the nodes in this typology, however one chooses to label them.

As a point of departure, I will assume the following definitions:

- (4)
- a. *Obligatory Control* (OC): The controller and the infinitive must be clausemates.
  - b. *Exhaustive Control* (EC): PRO must be identical to the controller.
  - c. *Partial Control* (PC): PRO must include the controller.
  - d. *Split Control*: Two matrix arguments jointly control (a plural) PRO.
  - e. *Non-Obligatory Control* (NOC): The infinitive need not have a clausemate controller.
  - f. *LD-control*: The controller and the infinitive are not clausemates.
  - g. *Arbitrary Control*: PRO has no argumental controller.
  - h. *Implicit Control*: The controller is not syntactically expressed.

Notice that the above terms apply to *types* of constructions, not to *tokens*, and there are some overlaps. In PC PRO must include the controller, but need not properly include it; thus some tokens of PC show identity between PRO and the controller, just like all tokens of EC do. Similarly, some tokens of NOC show local control, just like all tokens of OC do; and finally, in some tokens of OC or NOC the controller may be implicit.

The next question to ask is what constructions actually fall under each type; that is, how does the typology in (3) cut the pie of control constructions. Here real empirical issues arise; the boundary between OC and NOC, for example, is a topic of much controversy, and classificatory choices often depend on which grammatical properties are taken to be criterial and which peripheral. We will first take a look at this question in the domain of OC, and then proceed to NOC constructions.

### 3. EC and PC (CHAPTER 2)

It is a common observation that not all complement infinitivals are alike as far as control is concerned. The classical view, in fact, holds that cases like (5a) are fundamentally different from cases like (5b,c):

- (5) a. John tried (\*for Mary) to win the game.
- b. John wanted (for Mary) to win the game.
- c. John wondered how to win the game.

According to the classical view, (5a) is an instance of OC whereas (5b,c) are instances of NOC (Williams 1980, Chomsky 1981, Bresnan 1982, Manzini 1983, Bouchard 1984, Koster 1984, Martin 1996, Manzini & Roussou 1998, Wurmbrand 1998b, Hornstein 1999).<sup>1</sup> The basis for this distinction is the possibility of a *for*-complement in (5b) and the apparent arbitrary control in (5c), both of which are impossible in (5a).

There are good reasons to reject the classical view. Both (5b,c) display all the core properties of OC - barring LD-control, strict reading of PRO under ellipsis and *de re* interpretations (see discussion in chapter 2, section 3.2). As for the *for*-complement criterion, Manzini (1983) already observed that it does not reliably correlate with any other NOC property; selection for *for*-complement is governed by subtle lexical factors, not fully systematic, related to *emotivity* (see Kiparsky & Kiparsky 1970).

The notion that interrogative infinitival complements, as in (5c), license arbitrary control, also does not survive closer scrutiny. Notice that truly arbitrary PRO need not be linked to any grammatical antecedent:

- (6) a. John<sub>1</sub> thought that it was wrong [PRO<sub>arb</sub> to introduce him<sub>1</sub> to the dean].
- b. Sue<sub>1</sub> said that [PRO<sub>arb</sub> to buy her<sub>1</sub> nothing in Rome] would be unacceptable.

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<sup>1</sup> Not all of these authors treat (5b) and (5c) on a par, but they all classify at least one of them under NOC.

In contrast, PRO in a complement interrogative clause must always include a matrix controller; hence, the examples in (7) violate Condition B:<sup>2</sup>

- (7) a. \* John<sub>1</sub> wondered [who PRO<sub>1+</sub> to introduce him<sub>1</sub> to].  
 b. \* Sue<sub>1</sub> asked [what PRO<sub>1+</sub> to buy her<sub>1</sub> in Rome].

Our first conclusion, then, is that both (5b,c) belong to the OC category. More generally: *All* complement infinitives fall under OC. Why that must be so is taken up in chapter 3.

Nevertheless, there is something to the classical intuition that control in (5b,c) is somewhat more flexible than in (5a). This intuition is reconstructed in the present study through the distinction between EC and PC. Interrogative and *want*-type complements fall under PC, hence do not impose strict identity between the controller and PRO.

### 3.1. Properties of PC

Chapter 2 studies the EC/PC distinction across the whole range of infinitival complements, and reaches some non-trivial generalizations, possibly universal, about the distribution of each type. Before we state those generalizations, consider a few representative cases:

#### (8) *Exhaustive Control*

- a. The chair<sub>1</sub> managed [PRO<sub>1</sub> to gather the committee at 6].  
 b. \* The chair<sub>1</sub> managed [PRO<sub>1+</sub> to gather at 6].  
 c. Mary knew that John<sub>1</sub> began [PRO<sub>1</sub> to work (\*together) on the project].

#### (9) *Partial Control*

- a. The chair<sub>1</sub> preferred [PRO<sub>1+</sub> to gather at 6].  
 b. \* The chair<sub>1</sub> preferred [PRO<sub>1+</sub> to gather without him<sub>1</sub>].  
 c. Mary<sub>1</sub> thought that John<sub>2</sub> didn't know [where PRO<sub>1+2</sub> to go together].

*Manage* and *begin* are EC verbs, while *prefer* and *know* are PC verbs. Consider the minimal pair in (8a-b): The transitive *gather* is a distributive verb compatible with a singular subject, whereas the intransitive *gather* is a collective verb requiring a plural subject. Since *manage* is an EC verb, and the controller of PRO is singular, (8b) is ruled out. By contrast, the PC verb *prefer* can appear in exactly the same

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<sup>2</sup> I use the notation [DP<sub>1</sub>... [PRO<sub>1+</sub> ...]] to indicate partial control.

environment (9a).<sup>3</sup> That the latter is an instance of PC rather than NOC is witnessed in (9b), where Condition B imposes complete disjointness between PRO and the controller. The contrast between (8c) and (9c) illustrates again the EC/PC distinction, this time with the item *together* that requires a plural subject. Notice that no semantic anomaly is involved in the ungrammatical examples; the intended interpretations are always clear enough, yet the syntax makes them available only for a subclass of cases.

We break the domain of infinitival complements into seven subclasses, according to the semantic properties of the control verb: Aspectual (*begin, continue...*), modal (*need, able...*), implicative (*manage, dare...*), desiderative (*want, prefer...*), factive (*hate, regret...*), propositional (*claim, believe...*), interrogative (*wonder, ask...*). The partition is universal, although membership in each class is subject to language particular factors. For example, propositional infinitives are quite common in Romance but in English are only found with the verb *claim* (and maybe *pretend*). Likewise many aspectual and factive verbs in English appear with gerundive complements, although their control properties are unaffected.<sup>4</sup>

Testing all those verb classes in environments similar to (8)-(9) (see discussion in chapter 2, section 3.3), we arrive at the following dichotomy:

- (10) a. EC verbs are *implicative, aspectual or modal*.
- b. PC verbs are *factive, propositional, desiderative or interrogative*.

Generalizing yet to a higher level, one can observe that the feature distinguishing the EC type from the PC type is the presence of *tense* in the infinitive. Only PC complements may denote events that do not coincide with the matrix event (see chapter 2, section 4, for extensive discussion):

- (11) a. \*Yesterday, John managed to solve the problem tomorrow.
- b. Yesterday, John wanted to solve the problem tomorrow.

Following Stowell (1982) and Pesetsky (1991) we assume that infinitives have their own tense; however, unlike those authors, we maintain that this only applies to the PC class, the EC class (which is much smaller) consisting of tenseless infinitives. (10) then boils down to the generalization that an infinitival complement belongs to the PC class iff it is tensed.

Partial control has another interesting property which must be accounted for: It only induces *semantic* plurality on PRO, not *syntactic* plurality. This contrast is independent of control, as the following examples illustrate:

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<sup>3</sup> Some contextual salience of the group about to gather is needed to make (9a) felicitous, however no context can redeem (8b).

<sup>4</sup> Aspectual and modal verbs come in two guises – raising and control. The former is trivially expected to block PC, hence we concentrate on the latter.

- (12) a. I saw the committee gathering/dispersing.  
 b. I approve of the population acting together against the new regulations.  
 c. \* The government cleared themselves / each other of any responsibility.  
 d. \* I consider the delegation (to be) idiots.

Collective predicates and *together* can appear with subjects that are semantically plural but syntactically singular (12a,b); however, plural morphology on a predicate or an anaphor normally requires a syntactically plural subject (12c,d). Exactly what kind of plurality is required in specific contexts is subject to some lexical and dialectal differences; (12c,d), for example, are acceptable in British English. Limiting ourselves to those dialects of English where the distinction between semantic and syntactic plurality is robust, it is striking to see it at work in PC environments as well:

- (13) a. John told Mary that he preferred to meet at 6 today.  
 b. \* John told Mary that he preferred to meet each other at 6 today.
- (14) a. John told Mary that he didn't know which club to join together.  
 b. \* John told Mary that he didn't know which club to become members of.

As shown in chapter 2, section 3.4, this property of PC constructions is universal and purely morphological: Thus, languages in which *together* inflects for syntactic plurality disallow examples like (14a) while still allowing examples like (13a). Furthermore, the mismatch in semantic plurality allowed in PC is asymmetrical; a semantically plural controller cannot control a semantically singular PRO, even if the latter's reference is included in the former's:

- (15) a. The committee<sub>i</sub> was glad that the chair had agreed [PRO<sub>i</sub> to gather before the elections].  
 b. \* The chair<sub>i</sub> was glad that the committee had agreed [PRO<sub>i</sub> to wear a tie].

The empirical profile of PC is summarized in the generalization below:

(16) *The PC-Generalization*

In tensed complements, PRO inherits all phi-features from the controller, including semantic plurality, but not necessarily semantic singularity.

### 3.2. The Theory in a Nutshell

Our account makes certain assumptions about the grammatical expression of tense and phi-features (for the full analysis, see chapter 2, section 5). Following much work in generative grammar (for a recent formulation, see Rizzi 1997), I assume that the C(omp) head contains some tense information that has to be matched with the embedded T(ense) head. Following Pesetsky & Torrego (2000), I take this relation to be mediated by head movement; specifically, in tensed clauses T raises to C to check off an uninterpretable T-feature residing in C. I assume that T-to-C is unmotivated, hence blocked by economy, in untensed clauses.

Next, consider the status of semantic number. Unlike normal phi-features, semantic number is intrinsic to lexical nouns (e.g., *chair* is singular, *committee* is plural). Assuming PRO is no different, it can be selected from the lexicon either semantically plural [+SP] or semantically singular [-SP]. For functional heads like T-Agr or small *v*, we will assume that semantic number is an optional feature: They may either acquire it via agreement with a lexical DP, or lack it altogether. This may reflect the fact that [-SP] is a default value, non-distinct from [ØSP].

As for normal phi-features, assume that PRO is no different than any other noun, except for being an “anaphor” in the traditional sense: It freely picks phi-features before entering the derivation. In the case of PRO those features are anaphoric, but valued (like the phi-features of *himself*). Following Borer (1989), we also assume that the infinitival Agr is anaphoric and needs to be “identified”, although we implement “identification” not via Binding Theory but rather via Agree, as formalized in Chomsky (1998). Crucially, we will assume that PRO, being anaphoric, cannot value the anaphoric Agr by itself.

We now take OC to be an instance of Agree. The *probe* (i.e., attractor) is the matrix functional head that agrees with the controller: T for subject, small *v* for object and perhaps “applicative” *v* (Marantz 1993) for datives/obliques. The *goal* (i.e., attractee) is PRO or the infinitival Agr, the former activated in EC, the latter in PC. Schematic derivations are illustrated below (where DP is the controller and F is the matrix head that agrees with it):

- (17) a. [... F ... DP ... [<sub>CP</sub> [<sub>IP</sub> PRO T-Agr [<sub>VP</sub> t<sub>PRO</sub> ...] ] ] ] EC
- 
- Agree<sub>2</sub>                      Agree<sub>1</sub>
- Agree<sub>3</sub>
- b. [... F ... DP ... [<sub>CP</sub> T-Agr [<sub>IP</sub> PRO t<sub>T-Agr</sub> [<sub>VP</sub> t<sub>PRO</sub> ...] ] ] ] PC
- 
- Agree<sub>2</sub>                      Agree<sub>1</sub>
- Agree<sub>3</sub>

In (17a) the complement is untensed, so T-Agr remains in-situ. Agree<sub>1</sub> matches the phi-features of T-Agr and PRO (followed by raising of the latter to [Spec,TP]). The matrix head F enters two Agree relations - with the controller and with PRO. In virtue of Agree<sub>2</sub>, F inherits the semantic number of DP; in virtue of Agree<sub>3</sub>, PRO inherits it too, and we obtain the EC effect.

In (17b) the complement is tensed, so T-Agr raises to C, after Agree<sub>1</sub> took place. Being at the head of the CP phase, T-Agr escapes the “Phase Impenetrability Condition” (PIC) of Chomsky (1998, 1999), which makes elements below the head of a phase invisible to operations from outside. Agree<sub>3</sub> can then target T-Agr as a goal. If F is [+SP] (due to agreement with DP), T-Agr, which is picked from the lexicon without that feature, inherits [+SP] and feature matching requires PRO to be [+SP] as well; this explains the failure of a “downwards” shift in PC (cf. (15b)). However, if F is [-SP], T-Agr can remain [ØSP] since those two options are non-distinct. This allows PRO to be [+SP], explaining the PC effect.

Importantly, Agree cannot target T-Agr as a goal in (17a), since it is too low to be visible, by the PIC. Hence, PC is impossible in an untensed complement, where T-Agr remains in-situ. The fact that PRO in (17a) is visible implies the PIC should be relaxed to grant greater visibility to interpretable features - say, anywhere within the embedded phase, not just at its head/edge. The phi-features on PRO, although anaphoric, are valued, hence interpretable; those on T-Agr are uninterpretable (whether valued or not). This modification of the PIC seems plausible and inconsequential for standard cases of successive cyclicity. Notice that we assume, with Chomsky (1999), that uninterpretable features persist until the end of the superordinate phase before they erase. This makes the phi-features of T-Agr available for matrix operations, and enables those of F to enter two checking relations.

We thus derive the generalization in (16) and explain the internal patterns of control in the OC category. An important implication is that OC is established *derivationally*. In particular, in a substantial number of cases (i.e., in PC) semantic interpretation is affected by an uninterpretable element that does not survive to the LF interface, namely the infinitival Agr. A natural way to understand this is to make semantic interpretation cyclic, just like syntactic and phonological structure-building operations.<sup>5</sup>

#### 4. OC and NOC (CHAPTER 3)

Moving one layer up in the typology (3), and pursuing the same methodology as before, we now pose the following questions:

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<sup>5</sup> Further questions arise concerning the compatibility of subject control across an object with the Minimal Link Condition. In chapter 2, section 5.2, I argue that this phenomenon can be understood in terms of Richards's (1997) Principle of Minimal Compliance.

- (18) a. Distribution: *Where* do OC and NOC obtain?  
 b. Explanation: *Why* do OC and NOC obtain where they do?

Again, establishing the distributional generalizations is not a trivial matter. The debate over the proper demarcation of OC vs. NOC is tightly linked to the debate over the proper analysis of the so-called “Super-Equi” construction. Therefore, we will focus our attention on the latter, and by doing so will come to consider the interaction of control with extraposition and argument structure.

#### 4.1. *Super-Equi*

Super-Equi comes in two varieties - extraposition or intraposition (sentential subject). For a long time, since Grinder (1970), it was known that the choice of controller in these constructions is relatively free, although locality does govern a subclass of cases. The question was: which cases exactly? Grinder thought that a local controller (clausemate with the infinitive) can be skipped in intraposition (19a) but not in extraposition (19b):<sup>6</sup>

- (19) a. John said that making a fool of himself in public disturbed Sue.  
 b. \* John said that it disturbed Sue to make a fool of himself in public.

This position went unchallenged until Manzini (1983), who produced the following example of LD-control across an intervener in extraposition:

- (20) Mary knows that it would help Bill to behave herself in public.

Grinder’s analysis cannot account for (20), nor can Manzini’s account for (19b). In fact, no existing account of Super-Equi has ever resolved this empirical paradox (for an historical survey, see chapter 3, section 6).

The starting point of our discussion is the hypothesis that the crucial factor distinguishing (19b) from (20) is the *psychological* nature of the predicate governing the infinitive in the former (*disturb*), versus the non-psychological predicate in the latter (*help*). By that we simply mean that statements like “X disturbs Y” carry certain psychological entailments with respect to the mental state of Y, whereas statements like “X helps Y” carry no such entailments.

A minimal paradigm to illustrate this effect is the following:

- (21) a. Mary knew that it disturbed John [PRO to perjure himself / \*herself].  
 b. Mary knew that it damaged John [PRO to perjure himself / herself].

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<sup>6</sup> Actually, Grinder (1970) argued that even control of intraposed clauses is subject to his “Intervention Constraint”, however that constraint was formulated in a way which distinguished between (19a-b); see chapter 3, section 6 for discussion.

- c. Mary knew that [PRO perjuring himself / herself] disturbed John.
- d. Mary knew that [PRO perjuring himself / herself] damaged John.

The relevant generalizations are stated in (22):

- (22) a. In a structure [... X ... [it Aux Pred Y [<sub>S</sub> PRO to VP]]], where Y and S are arguments of Pred:
  - i) If Pred is psychological, Y must control PRO.
  - ii) If Pred is non-psychological, either X or Y may control PRO.
- b. In a structure [... X ... [<sub>S</sub> [<sub>S</sub> PRO to VP] Pred... Y]], either X or Y may control PRO.

In short, LD-control is susceptible to intervention only in extraposition, and only when the intervener bears an EXPERIENCER  $\theta$ -role. The implication for question (18a) is that among Super-Equi constructions, only extraposition under a psychological predicate falls under OC; all other cases fall under NOC. In chapter 3, section 1.2, I show that this generalization holds not only in English but also in German, Italian, Spanish, Swedish and Hebrew.

This remarkable state of affairs calls for explanation, an explanation that will, inevitably, hinge on both the argument structure of the matrix predicate and the syntactic position of the infinitive. Let us sketch the analysis we develop in chapter 3, section 2.

The first and most important ingredient is a structural constraint on the operation of OC:

(23) *The OC Generalization*

In a configuration [... DP<sub>1</sub> ... Pred ... [<sub>S</sub> PRO<sub>1</sub> ... ] ...], where DP controls PRO: If, at LF, S occupies a complement/specifier position in the VP-shell of Pred, then DP (or its trace) also occupies a complement/specifier position in that VP-shell.

(23) is the present analogue of the distinction between open and closed grammatical functions in Bresnan (1982), or that between governed and ungoverned positions in Manzini (1983). (23) amounts to the claim that any VP-internal infinitive must find a VP-internal controller, where VP is understood in the broad sense as the projection hosting all arguments (including the external one). It follows that all VP-external infinitives fall under NOC. Notice that *which* argument inside VP is selected as the controller is not dictated by (23): In chapter 5 we present extensive evidence that controller choice in OC is a semantic/pragmatic issue.

The second ingredient is a PF-constraint on extraposition:

(24) *Extraposition*

VP-internal clauses must be peripheral at PF.

(24) corresponds to the crosslinguistic observation that embedded clauses are typically peripheral to the VP and seldom intervene between a predicate and other internal arguments. Where this does not obtain in base-structure (because of thematic mapping), the embedded clause must extrapose. I assume that extraposition in this case is adjunction to VP, and (as is standardly assumed) an adjunct is not dominated by its host category.

The third ingredient concerns the interpretation of chains:

(25) *Chain Interpretation*

Any link in a chain may be the LF-visible link.

(25) is the null hypothesis under the view that traces of movement are full copies of the element moved (Chomsky 1995). Reconstruction effects that track the path of movement suggest that, in principle, any link along that path can be the one feeding the interpretive component (Fox 1999, Bobaljik 1999), at least for A-bar movement. Assuming that adjunction falls under that type, extraposition yields two copies (base and extraposed). The extraposed copy must feed PF to satisfy (24), however either copy may feed LF. “Reconstruction” is a name for the situation where the higher copy is pronounced and the base copy is interpreted. Since the base position is VP-internal and the extraposed one is VP-external, interpreting the former results in OC, while interpreting the latter results in NOC, assuming (23).

The final ingredient concerns the hierarchical projection of arguments:

(26) *Argument Projection*

- a. EXPERIENCER is generated above CAUSER.
- b. CAUSER is generated above GOAL/PATIENT/THEME.

(26a) has been invoked to explain several unaccusative properties of psych-verbs (Belletti & Rizzi 1988), as well as certain scope reversal properties they display (Pesetsky 1995, Stroik 1996). (26b) is a standard assumption that needs no further comment. Notice that the two clauses of (26) never apply jointly; no predicate selects EXPERIENCER, CAUSER and GOAL/PATIENT at the same time.

This system derives the desired results, under the assumption that movement is constrained by economy. Consider the following partial representations (abstracting away from V-movement), where the subscript “1” indicates local control and the subscript “2” non-local (LD or arbitrary) control:

- (27) a. [<sub>VP</sub> John<sub>1</sub> [<sub>V</sub> please [<sub>S</sub> PRO<sub>1/\*2</sub> to work late]<sub>PF/LF</sub>]]  
 b. \* [[<sub>VP</sub> John<sub>1</sub> [<sub>V</sub> please [<sub>S</sub> PRO<sub>1/\*2</sub> to work late] ] ] [<sub>S</sub> PRO<sub>1/\*2</sub> to work late]]
- (28) a. [[<sub>VP</sub> [<sub>S</sub> PRO<sub>1/\*2</sub> to work late]<sub>LF</sub> [<sub>V</sub> help John<sub>1</sub>]] [<sub>S</sub> PRO<sub>1/\*2</sub> to work late]<sub>PF</sub>]  
 b. [[<sub>VP</sub> [<sub>S</sub> ~~PRO<sub>1/\*2</sub> to work late~~] [<sub>V</sub> help John<sub>1</sub>]] [<sub>S</sub> PRO<sub>1/2</sub> to work late]<sub>PF/LF</sub>]

- (29) a.  $[_{IP}[_S \text{ PRO}_{1/2} \text{ to work late}]_{PF/LF} \text{ would } [_{VP} \text{ John}_1[_{V'} \text{ please } [_S \text{ PRO}_{1/*2} \text{ to work late}]]]]]$   
 b.  $[_{IP}[_S \text{ PRO}_{1/2} \text{ to work late}]_{PF/LF} \text{ would } [_{VP} [_S \text{ PRO}_{1/*2} \text{ to work late}] [_{V'} \text{ help John}_1]]]$

Under a psychological predicate, the CAUSER infinitive is VP-peripheral to begin with (27a). Extraposition is unmotivated, regardless of PF/LF-copy resolution, hence blocked by economy (27b). Since the infinitive stays in-situ, OC applies and control must be local.

Under a non-psychological predicate, the CAUSER infinitive is generated above the other object, hence is not VP-peripheral (28). Extraposition applies, and LF may interpret either the base copy (28a) or the higher one (28b); the latter option gives rise to NOC, since the infinitive occupies a VP-external position at LF.

In intraposition (29), the infinitive raises to [Spec,IP] to satisfy EPP. This is an instance of A-movement, and we may assume (following Chomsky 1995) that unless special circumstances demand otherwise, it is the higher link of the chain that gets interpreted. Again the infinitive escapes the domain of OC, this time regardless of the semantic class of the matrix predicate.

We thus derive the generalizations in (22), resolving the empirical paradox raised by (19b) and (20). We must still explain the OC Generalization (23), which is doing the right work for us but is so far no more than a distributional statement. Before we do so, let us turn to some striking confirmation for the analysis just sketched.

The interaction of our assumptions about OC and extraposition yields the following consequence: Whenever the infinitive is (syntactically) in situ, control is local; and whenever control is non-local (arbitrary or long-distance), the infinitive is extraposed to a position adjoined to VP. Assuming that adjuncts are islands to extraction, we expect a correlation between non-local control and failure of extraction.

The prediction is borne out, both for arbitrary and for LD-control (for more evidence, see chapter 3, section 3):

- (30) a. It would kill the workers<sub>1</sub> [PRO<sub>1</sub> to build this dam].  
 b. What<sub>2</sub> would it kill the workers<sub>1</sub> [PRO<sub>1</sub> to build t<sub>2</sub> ?]  
 c. It would kill the forest [PRO<sub>arb</sub> to build this dam].  
 d. \* What<sub>2</sub> would it kill the forest [PRO<sub>arb</sub> to build t<sub>2</sub> ?]
- (31) a. Hillary thinks it damaged Bill<sub>1</sub> [PRO<sub>1</sub> to talk about himself on the Dave Letterman show].  
 b. That's the talk show<sub>2</sub> that Hillary thinks it damaged Bill<sub>1</sub> [PRO<sub>1</sub> to talk about himself on t<sub>2</sub> ].  
 c. Hillary<sub>1</sub> thinks it damaged Bill [PRO<sub>1</sub> to talk about herself on the Dave Letterman show].

- d. \* That's the talk show<sub>2</sub> that Hillary<sub>1</sub> thinks it damaged Bill [PRO<sub>1</sub> to talk about herself on t<sub>2</sub>].

Notice that both *kill* and *damage* are non-psychological predicates, allowing either OC (the (a) examples) or NOC (the (c) examples). However, when extraction takes place from the infinitive, NOC is no longer available ((b) versus (d)). This is because extraction requires the LF-copy of the infinitive to be the base one, whereas NOC requires it to be the extraposed one; those conflicting demands cannot be simultaneously satisfied.

Evidence from Weak Crossover converges on the conclusion that OC and NOC in extraposition, although string-identical at PF, involve different structures:

- (32) a. It would ruin this tune<sub>1</sub> [PRO<sub>arb</sub> to play it<sub>1</sub> on a flute].  
 b.?? What tune<sub>1</sub> would it ruin t<sub>1</sub> [PRO<sub>arb</sub> to play it<sub>1</sub> on a flute]?  
 c. What tune<sub>1</sub> would it ruin t<sub>1</sub> [PRO<sub>arb</sub> to play e<sub>1</sub> on a flute]?

Since the matrix object does not c-command the infinitive when arbitrary control obtains, the configuration in (32b) violates Weak Crossover. Predictably, the sentence becomes acceptable if the pronoun is replaced by an empty category, as in (32c); this involves a parasitic gap strategy, which is licensed precisely because the “anti c-command” condition is satisfied.

#### 4.2. Anaphors and Logophors

The rationale behind the OC Generalization (23) is already implicit in the proposal, made in chapter 2, that OC involves an Agree relation between the matrix functional head F that agrees with the controller and PRO/T-Agr. Agree, which underlies standard chain formation, has certain locality properties, which are indeed observed in OC. In particular, it obeys the “Phase Impenetrability Condition” (see above); thus the head F (and its specifier, the controller) must be found within the phase immediately dominating the infinitival CP phase. The locality of OC follows. The fact that the controller must be a complement or specifier in the matrix VP-shell reflects the fact that only those positions enter into Agree relations with the functional heads that head the layers of the shell.

The last ingredient of (23) to explain is why the locality conditions only apply when the infinitive is VP-internal (complement or specifier):

- (33) a. Infinitive is VP-complement/VP-specifier ⇒ OC  
 b. Infinitive is IP-subject / adjunct ⇒ NOC

We have already seen that this partitioning is supported by extraction asymmetries: OC infinitives are transparent to extraction, whereas NOC infinitives - like subjects and adjuncts in general - are islands. I argue that the *same* structural distinction is

How is PRO interpreted in an island environment, that is, when it can enter no syntactic relation with elements outside the island? Following Reinhart & Reuland (1993), I suggest that an anaphoric element which fails to be syntactically licensed is interpreted as a *logophor*. Logophors are licensed by discourse factors such as focus, perspective and center of consciousness or communication. Indeed, it is an old insight in generative grammar that LD-control in Super-Equi has a lot in common with “picture-anaphora” (Grinder 1970, Kuno 1975, Lebeaux 1985). This view contrasts with an alternative view, by which PRO in LD-control is pronominal (Bouchard 1984, Hornstein 1999). Since logophors are subject to more stringent antecedence conditions than pronouns, the two views make different predictions.

(34) a.  $[(\text{his}_i) \text{ having shaved already}]$  shows that Mary arrived more than 5 minutes after John<sub>i</sub> did.

b. John's<sub>i</sub> friends think it is illegal [ $^{*}(\text{for him})$  to feed himself].

We conclude that PRO in NOC is a logophor rather than a (silent) pronoun. This property emerges in island environments, where the infinitival Agr fails to enter an Agree relation with an element outside the island and cannot be syntactically identified.

So far we have concentrated on questions (2a,c,d) - those that establish the typology of control. I now turn to the subject matter of question (2e), the interpretation of PRO. There are many aspects to this question, some of which we leave untouched (e.g., the *de re/de se* distinction, deictic or generic PRO etc.). The contribution of this study to the semantics of control is in providing some new arguments which bear on the classical debate between the *predicational* and the *propositional* approaches to OC.

The predication approach holds that infinitives in OC denote unsaturated predicates (properties). The open position corresponds to the subject of the infinitive, which under some versions is projected as PRO and under others is completely missing from the syntax (the latter take controlled infinitives to be bare VP's). If PRO is present, then it is merely a lambda-variable. The semantic procedure

associates the controller with the infinitive by predicating the latter of the former (see Bach 1979, 1982; Williams 1980; Dowty 1985; Lebeaux 1985; Chierchia 1984, 1989, 1990; Clark 1990).

The propositional approach holds that infinitives in OC denote closed propositions, just like finite clauses do. PRO is projected in the syntax and saturates the subject position of the infinitive. The semantic procedure associates the controller with PRO via an anaphoric relation of sorts (see Chomsky 1980, 1981; Koster & May 1982; Manzini 1983; Bouchard 1984; Koster 1984; Borer 1989).

Notice that the issue under debate is not the existence of PRO; both approaches are compatible with a syntactic PRO. Rather, at stake is the manner in which PRO composes with the infinitival VP. This determines the semantic type of the infinitive, and the question is: Is the infinitive a predicate or a proposition? I will argue that there are good reasons to adopt the propositional approach. We will see that within a particular empirical domain - adjectival complementation - that approach can express natural generalizations that the predication approach fails to.

The starting point to our discussion is an observation made by Faraci (1974), which can be generalized and stated in the following way:<sup>7</sup>

- (35) Given a predicate P that takes an infinitival complement C:
- a. If P is psychological, C contains at most one bound gap - in subject position.
  - b. If P is non-psychological, C contains exactly one bound gap (subject or object).

(35) points to a dual asymmetry between complements to psych-adjectives and complements to non-psych adjectives - both in the *necessity* and in the *position* of the gap. The following paradigm illustrates these asymmetries (*ready<sub>P</sub>* stands for the psychological sense of *ready*, *ready<sub>M</sub>* stands for its non-psychological, "material" sense):

- (36) a. The patient is *ready<sub>P</sub>*/anxious/eager for the doctor to operate on him.  
 b. \* The patient is *ready<sub>P</sub>*/anxious/eager for the doctor to operate on \_\_\_\_.  
 c. The tumor is *ready<sub>M</sub>* for the doctor to operate on \_\_\_\_.  
 d. \* The tumor is *ready<sub>M</sub>* for the doctor to operate on it.

[Faraci 1974: 80-82]

- (37) a. Mary<sub>i</sub> was happy [*e<sub>i</sub>* to assist anyone].  
 b. Mary<sub>i</sub> was happy [*e<sub>i</sub>* to be assisted].  
 c. \* Mary<sub>i</sub> was happy [*e<sub>arb</sub>* to assist *e<sub>i</sub>*].

- (38) a. The book<sub>i</sub> is available [*e<sub>i</sub>* to be read].

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<sup>7</sup> *Tough*-adjectives, although mostly psychological, do not fall under (35a); see chapter 4, section 4.2 for discussion.

- b. \* The book<sub>i</sub> is available [<sub>e<sub>arb</sub></sub> to read it<sub>i</sub>].  
 c. The book<sub>i</sub> is available [<sub>e<sub>arb</sub></sub> to read e<sub>i</sub>].

I propose that a psych-adjective denotes a two place relation between an individual (*experiencer*) and an eventuality (*target/subject-matter*). The eventuality is of a propositional type, and may be expressed by a gapless infinitive (36a) or a subject-gap one (37a,b). A non-psych adjective, by contrast, denotes a one place property of an individual (*theme*). That property, in turn, can be modified by a *predicative* infinitival. The unsaturated position of that infinitive may be either the object (36c)/(38c) or the subject (38a).

In other words, we assume that the syntax-semantics mapping makes available three types of infinitives that are mapped onto two semantic types:

(39) *Semantic Typology of Infinitival Clauses*

- a. [<sub>CP</sub> Op<sub>i</sub>[<sub>IP</sub> DP/PRO<sub>arb</sub>...[<sub>VP</sub> ... t<sub>i</sub>]]] ⇒ CP denotes a predicate (<e,<s,t>>)  
 b. [<sub>CP</sub> Op<sub>i</sub>[<sub>IP</sub> t<sub>i</sub>... [<sub>VP</sub> ... ]]] ⇒ CP denotes a predicate (<e,<s,t>>)  
 c. [<sub>CP</sub> [<sub>IP</sub> DP/PRO...[<sub>VP</sub> ... ]]] ⇒ CP denotes a proposition (<s,t>)

Since a psych-adjective combines with a propositional infinitive, which can only be formed as in (39c) - we derive (35a); since a non-psych adjective combines with (i.e., is modified by) a predicative infinitive, which can be formed either as in (39a) or (39b) - we derive (35b).

The non-trivial part of this analysis is the distinction between (39b-c); that is, we claim that a subject-gap infinitive may be propositional or predicative, depending on context; moreover, in the former case it is an *argument* of the (psych) adjective, whereas in the latter it is a *modifier* of the (non-psych) adjective.

This distinction receives independent support from familiar argument/modifier asymmetries. We give a sample of them here (see chapter 4, section 3 for discussion).

A psych adjective may occur without an infinitival complement only if the latter is explicit in the context (i.e., ellipsis), whereas a non-psych adjective allows free complement omission:

- (40) a. A: Someone should do the laundry, you know.  
       B: Well, I'm willing (to do the laundry).  
       b. A: The laundry is really dirty, you know.  
       B: Well, I'm willing \*(to do the laundry).  
       B': Well, the washing machine is available (to do the laundry).

This contrast is explained if the infinitive is an argument of the psych-adjective but a modifier of the non-psych adjective.

Second, it is well-known that arguments are transparent to extraction but non-arguments - namely, adjuncts and modifiers - are opaque. Indeed, extraction is possible from a complement of a psych-adjective but not from that of a non-psych-adjective (concentrate below on the downstairs reading of the *wh*-word):

- (41) a. When<sub>i</sub> is John ready<sub>P</sub> to test the car t<sub>i</sub>? (answer: tomorrow)  
 b. How<sub>i</sub> is John ready<sub>P</sub> to eat the dish t<sub>i</sub>? (answer: without chopsticks)
- (42) a. \* When<sub>i</sub> is the car ready<sub>M</sub> to test t<sub>i</sub>? (answer: tomorrow)  
 b. \* When<sub>i</sub> is the car ready<sub>M</sub> to be tested t<sub>i</sub>? (answer: tomorrow)  
 c. \* How<sub>i</sub> is the dish ready<sub>M</sub> to eat t<sub>i</sub>? (answer: with chopsticks)  
 d. \* How<sub>i</sub> is the dish ready<sub>M</sub> to be eaten t<sub>i</sub>? (answer: with chopsticks)

Third, in VP-fronting configuration, the infinitive must be adjacent to a non-psych-adjective but not to a psych-adjective:

- (43) a. Ready<sub>P</sub>/anxious/willing to serve (his country) though I'm sure John is,  
 I doubt that he is anxious to be posted in Bosnia.  
 b. Ready<sub>P</sub>/anxious/willing though I'm sure John is to serve (his country),  
 I doubt that he is anxious to be posted in Bosnia.
- (44) a. Ready<sub>M</sub> to serve (be served) though I'm sure the soup is,  
 I doubt that the guests will want it on such a hot day.  
 b.?? Ready<sub>M</sub> though I'm sure the soup is to serve (be served),  
 I doubt that the guests will want it on such a hot day.

Fourth, arguments but not modifiers can strand prepositions. Notice that both psych and non-psych adjectives can freely take *for*-PP's:

- (45) a. John is ready<sub>P</sub> for the exam.  
 b. The soup is ready<sub>M</sub> for dinner.

However, a *what*-question can be answered with an infinitive only in the psych-variant; the non-psych variant is marginal at best:

- (46) a. John is ready<sub>P</sub> to take the exam.  
 b. What is John is ready<sub>P</sub> for? To take the exam.  
 c. The soup is ready<sub>M</sub> to eat / be eaten.  
 b.?? What is the soup ready<sub>M</sub> for? To eat / be eaten.

All these tests demonstrate that the infinitival complement of a psych-adjective functions as an argument, whereas that of a non-psych adjective functions as a modifier.

Recall that the modifier infinitive is semantically a predicate, formed by null operator movement (47a); in this respect, it is formally equivalent to another type of predicative subject-gap infinitive - namely, a relative clause of the sort in (47b):

- (47) a. The book<sub>1</sub> is available [<sub>CP</sub> Op<sub>1</sub> [ t<sub>1</sub> to be read]].  
 b. The right person<sub>1</sub> [<sub>CP</sub> Op<sub>1</sub> [ t<sub>1</sub> to fix the sink]] could not be found.

Following Clark (1990), I assume that null operator chains are subject to certain case requirements. Whether these requirements are met in the subject position of infinitives is a language-specific matter. Nevertheless, given the formal equivalence of the infinitives in (47a,b), other things being equal, our analysis makes the following non-trivial crosslinguistic prediction:

(48) *A Crosslinguistic Correlation*

A language has subject infinitival relatives iff it has subject-gap infinitival complements to non-psych adjectives.

I have tested 11 languages for this striking prediction (see chapter 4, section 5). Languages that have both constructions are English, Italian, Québec French, Brazilian Portuguese, Spanish, Swedish and the German dialect of Itzgründisch. Languages that lack both constructions are Upper Austrian, Russian, Hebrew and Hindi. I have not found a language that has one of the two constructions mentioned in (48) but not the other. This is quite remarkable, given the superficial disparity between them.

To conclude, we have constructed three arguments for the propositional approach to control and against the predicational one. The first argument concerns the account for the generalizations in (35). The clausal typology offered in (39) treats PRO-infinitivals and *for*-infinitivals on a par, both being propositional; we thus understand why both fall under a unitary statement, namely (35a). The predicational approach would be hard-pressed to explain why subject gap infinitives figure in (35a) but object gap infinitives do not; if both are always predicative, the subject-object asymmetry is a mystery.

Second, the predicational approach would have to stipulate, in light of the argument/modifier asymmetries noted above, that a predicative infinitival can be either an argument or a modifier; whether that distinction is semantically meaningful is not obvious under this view, since the infinitive itself has the same denotation. That would fail to capture the observation that at least some argumenthood tests (e.g., extraposition and preposition stranding) *are* sensitive to semantic interpretation. By contrast, the propositional approach preserves a tight mapping

between the syntactic status of an infinitive as an argument or an adjunct and its semantic type.

Third, the crosslinguistic correlation in (48) is fully consistent with the propositional view but quite puzzling under the predication view. If both control and relativization involve predicating an infinitive of an NP, why is the former (apparently) a universal option whereas the latter is parametric? Moreover, we saw that the availability of subject gap complements to non-psych adjectives does *not* correlate with that of subject gap complements to psych adjectives; again, if the same mechanism underlies both constructions, this disparity is mysterious. However, if the licensing of PRO and the licensing of an A-bar variable in the subject of an infinitive are subject to different conditions, then the empirical pattern is not surprising.

## 6. IMPLICIT CONTROL AND CONTROL SHIFT (CHAPTER 5)

Finally, we turn to the subject matter of question (2f) - the choice of controller. This topic is quite complex; despite the impressive literature on it, basic issues are still unresolved. This state of affairs is partly due to the quality of the data, which is often not sharp enough to decide between competing analyses. My purpose in chapter 5 of this book is not to advance any novel account of the data, but rather to sort them out in a way which will hopefully be useful for future research. That is, we are after the empirical generalizations that underlie the phenomena of implicit control and control shift.

### 6.1. *Implicit Control*

We most frequently find implicit control with agents of passives and with datives. That implicit datives/benefactives can control is an old observation, going back to Kimball (1971). Bresnan (1982) cites (49a); Epstein (1984) argues that (49b) is not a case of arbitrary control but rather OC by an implicit argument, itself bound by a universal quantifier (49c); and Koster (1984) similarly argues that the apparent LD-control in (49d) is in fact OC by an implicit experiencer, as witnessed by the intervention effect in (49e):

- (49) a. Louise gestured/said/signaled (to Tom<sub>i</sub>) [PRO<sub>i</sub> to follow her].
- b. It is fun [PRO to eat ice-cream].
- c.  $\forall x$  [it is fun **for** **x** [for **x** to eat ice-cream]]
- d. Mary<sub>i</sub> said it was difficult [PRO<sub>i</sub> to take another topic].
- e. Mary said it was difficult for Bill<sub>i</sub> [PRO<sub>i</sub> to take another topic].

At times it has been suggested that the *type* of control exerted by implicit arguments is different than the type of control exerted by overt ones (Bresnan 1982, Bouchard 1984, Huang 1989, Chierchia 1989), perhaps even falling under NOC. In fact, there

is no reason to believe that implicit control (by internal arguments) is any different from normal OC (see chapter 5, section 1.2, for extensive discussion).

An independent issue which is sometimes conflated with implicit control is the question of the syntactic expression of arguments. Which arguments must be expressed and which may be left implicit is an interesting question; both lexical information and language particular factors seem to affect that decision. For example, it appears that direct objects are rarely omissible, whereas dative arguments can drop to varying degrees across languages (in English, only with communication verbs; in Hebrew, also with *recommend*, *permit*, etc.). Crucially, though, the issue is orthogonal to control; arguments that are (not) omissible in non-control contexts are also (not) omissible as controllers:

- (50) a. John convinced \*(Mary) of his innocence.
- b. John convinced \*(Mary) [PRO to believe him].
- c. Mary forced the song \*(on John).
- d. Mary forced \*(John) [PRO to sing].
  
- (51) a. John said the secret (to Mary).
- b. John said (to Mary) [PRO to listen to him].
- c. Mary helped (John) with the dishes.
- d. Mary helped (John) [PRO (to) do the dishes].

Although some datives can drop, not all oblique controllers can drop (e.g., *Tom pleaded \*(with Anne) to understand his situation*). This suggests that the line between OC and NOC is not sensitive to (unrestricted vs. restricted) grammatical functions, contra Bresnan (1982).

Turning to control by implicit agents, the picture becomes considerably more complex. Implicit agents can control into complements of impersonal passive constructions and into adjuncts:

- (52) a. It was decided to leave.
- b. Er werd geprobeerd [(om) PRO Bill te bezoeken].  
there was tried (Comp) PRO Bill to visit  
Lit: 'It was tried to visit Bill'

[Dutch: Koster 1984]

- (53) a. The game was played wearing no shoes.
- b. The president was elected without considering his competence.

[Roeper 1987]

The adjunct case seems to involve certain restrictions on the animacy and "topichood" of the implicit agent (see Kawasaki 1993). Consider the following paradigm:

- (54) a. [Before PRO entering the basement], the rain washed the stairs.  
 b. [Before PRO entering the basement], the stairs were washed.  
 c. [Before PRO entering the basement], Mary was washed.

Those speakers who accept (54b) construe the implicit controller as human - even though nothing about the pragmatics of the situation requires that (cf. (54a)). Still, a human subject, as in (54c), overrides the implicit agent and must be the controller itself. When the antecedent of PRO is salient enough, it need not be grammatically represented:

- (55) After pitching the tents, darkness fell quickly. [Kawasaki 1993]

The only case where an implicit agent is blocked from controlling is in the passive of *promise*:

- (56) \* John was promised (by Mary) to leave.

This case was claimed to illustrate “Visser’s Generalization” - subject control verbs do not passivize. In fact, as I show in chapter 5, section 2, it is not clear that the “generalization” covers more than this single case. Two basic questions arise: i) Why is an implicit agent different than an implicit dative as an OC controller? ii) Why can the implicit agent control in impersonal passives and adjuncts, but not in (56)? There have been quite a few attempts to explain Visser’s Generalization, but as far as I know, none of them provide satisfactory answers to those two questions (see Bach 1979, Chomsky 1980, Williams 1980, Bresnan 1982, Ružička 1983, Koster 1984, Farkas 1988, Larson 1991).

Another conundrum associated with implicit agent control is the behavior of rationale clauses. Consider the following paradigm:<sup>8</sup>

- (57) a. The boat was sunk [PRO to collect the insurance].  
 b. \* The boat sank [PRO to collect the insurance].  
 c. \* The ship sank (in order) for the owner to collect the insurance.  
 d. \* The ship was sunk to become a hero.  
 e. Grass is green to promote photosynthesis.  
 f. The shopwindow has a big sale sign in it (in order) to attract customers.  
 g. \* Marijuana was smoked to become illegal in the 1930s.

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<sup>8</sup> The examples in (57) are collected from Williams (1974, 1985), Manzini (1986), Roeper (1987), Lasnik (1988), Farkas (1988), Clark (1990) and Kawasaki (1993).

There are three major proposals in the literature as to the identity of the controller of PRO in rationale clauses:

- (58) a. The controller is the matrix (possibly implicit) agent.
- b. The controller is the purposeful causer of the matrix event.
- c. The controller is the matrix event.

In favor of (58a) is the familiar contrast between the passive and the unaccusative cases (57a-b). However, the ungrammaticality of (57c), without PRO, suggests that the unaccusative case (57b) is ruled out independently of control. Moreover, if a matrix agent is all that is necessary, then (57d) is also expected to be acceptable, contrary to fact. Finally, examples like (57e,f), where there is no matrix agent, argue conclusively against (58a). One then might take (57e,f) as evidence for (58b). However, a purposeful causer should be available in (57d) as well. Option (58c) seems to correctly rule in (57e,f) and rule out (57d) (events cannot become heroes); but it cannot extend to (57a) (events cannot collect insurance), nor can it readily rule out (57g) (smoking marijuana can become illegal).

Overall, then, it seems that none of the options in (58) is fully explanatory, although (I suspect) (58b) is essentially correct, with an unexplained residue, namely that rationale clauses under matrix passives must be agentive.

## 6.2. Control Shift

Control shift refers to a situation where the controller shifts from one matrix argument (the unmarked controller) to another (the marked controller). Two types of shifts are usually attested: Agent --> Goal and Goal/Theme --> Agent. They are illustrated in (59) and (60), respectively:

- (59) a. John<sub>i</sub> was promised [PRO<sub>i</sub> to be allowed to leave].
- b. Grandpa promised the children<sub>i</sub> [PRO<sub>i</sub> to be able to stay up for the late show].
- (60) a. Susie<sub>i</sub> persuaded the teacher [PRO<sub>i</sub> to be allowed to leave early].
- b. John<sub>i</sub> begged Mary [PRO<sub>i</sub> to be allowed to consult a doctor].

Judgments vary as to the degree of acceptability of specific examples, and pragmatic factors are relevant as well. Thus, with a verb like *ask*, all that is needed to shift the controller is some indication of the authority relations:

- (61) a. The pupil asked the teacher to leave early.
- b. The guard asked the prisoner to leave the room.

[Farkas 1988]

Normally, some modal force, explicit or not, is present in the infinitive under control shift. In English, *be allowed to* complements achieve that goal. In other languages, like Hebrew, a modal reading is not necessary, and embedded copular/passive predicates can trigger the shift from agent to goal control:

- (62) ha-menahel hivti<sub>ax</sub> li [PRO<sub>i</sub> lihiyot ha-/lehitmanot la-  
the-manager promised to-me PRO to be the-/to be appointed to-the-  
ozer šelo].  
assistant his  
'The manager promised me<sub>i</sub> [PRO<sub>i</sub> to be (appointed as) his assistant]'

A common insight in the literature is that these examples involve not just a shift in the controller but also a more global shift in the event structure of the clause. The analyses pursued invoke notions like thematic binding (Melvold 1985), intentional causer (Farkas 1988), semantic coercion (Sag & Pollard 1991) and deontic authority (Petter 1998) to capture the semantic relation persisting between the *unmarked* controller and the infinitive even after the shift to the marked controller. In chapter 5, section 3.2, I try to assess the merits and faults of those proposals, concluding that the semantic relation implicated in these constructions is quite abstract and not grammaticalized in any unique way.

This discussion bears directly on the status of theories attempting to deduce the identity of the controller solely on the basis of syntactic locality. Those descendants of the Minimal Distance Principle (MDP), due to Rosenbaum (1967), include Larson (1991), Martin (1996), Hornstein (1999) and Manzini & Roussou (2000). Analyzing Larson's proposal in detail, I show in chapter 5, section 4, that it fails to capture the real generalization governing subject control. In particular, I show that the analysis based on dative shift fails to extend beyond the single verb *promise* (*vow* and *threaten* are immediate counterexamples), while making incorrect predictions w.r.t. double object verbs that exhibit *object* control (e.g., *allow*). Overall, the MDP is untenable, and the question of controller choice (inside the OC domain) is best handed over to semantic/pragmatic considerations.

## 7. THE OVERALL PICTURE AND OPEN ISSUES

Putting together the different types and mechanisms of control discussed in this book, we arrive at the following picture:

(63) A Typology of Control

| <b>Obligatory Control</b>                     |                                             | <b>Non-Obligatory Control</b>                |                                                             |
|-----------------------------------------------|---------------------------------------------|----------------------------------------------|-------------------------------------------------------------|
| <i>Distribution:</i> VP-internal infinitives  |                                             | <i>Distribution:</i> VP-external infinitives |                                                             |
| <i>Mechanism:</i> <u>Agree</u>                |                                             | <b>LD-Control</b>                            | <b>Arb-Control</b>                                          |
| <b>Exhaustive Control</b>                     | <b>Partial Control</b>                      | <i>Mechanism:</i><br><br>PRO is a logophor   | <i>Mechanism:</i><br><br>PRO is bound by a generic operator |
| <i>Distribution:</i><br>Tenseless infinitives | <i>Distribution:</i><br>Tensed infinitives  |                                              |                                                             |
| <i>Mechanism:</i><br><u>Agree</u> (F,PRO)     | <i>Mechanism:</i><br><u>Agree</u> (F,T-Agr) |                                              |                                                             |

Looking at this table, one may conclude that there is no such thing as “the theory of control”. Instead, there is a cluster of subtheories, each explaining a particular aspect of control phenomena, in particular contexts. This may seem a rather dispiriting conclusion for a study aiming to decompose control into its elements; decomposition went so far that nothing was “left” that is intrinsic to our subject matter. An uplifting perspective, however, is also available: We now understand control better precisely because so much of it reduces to well-understood principles operating elsewhere in the grammar. I leave it to the reader to choose between those alternative conclusions.

For the sake of completeness, it should be noted that some control constructions are left out of the table in (63). These include control into adjuncts - rationale/purpose clauses, temporal adverbials, absolutive adjuncts etc.; “VP-external infinitives” in the NOC rubric of table (63) are restricted to extraposed and intraposed (argumental) clauses. We also offer virtually no discussion of control into nominals. As to adjunct control, in all likelihood this is not a homogeneous phenomenon, e.g., rationale clauses and temporal adverbials have radically different control properties, some of which lie outside sentence grammar. Part of the complexity that arises in this domain is discussed in chapter 5, section 2; however, a full, systematic treatment of these cases is beyond the scope of this study. As far as I know, the present analysis is not inconsistent with the facts we have left outside, and may have natural extensions to account for them as well.